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Binary and Gray Code Conversion with Step-by-Step Examples

Binary to Gray Code :

Rule: Copy the first binary digit. For each next digit: if current binary and previous binary are the same → 0, otherwise → 1.

Step	Binary Digits	Gray Result
Start	Binary = 1011	_
1	First digit = 1	Gray = 1
2	Compare (1,0) → different	Gray = 11
3	Compare (0,1) → different	Gray = 111
4	Compare (1,1) → same	Gray = 1110

Gray Code to Binary :

Rule: Copy the first Gray digit. For each next digit: if Gray=0 → copy the previous Binary digit, if Gray=1 → take the opposite of the previous Binary digit.

Step	Gray Digits	Binary Result
Start	Gray = 1110	_
1	First digit = 1	Binary = 1
2	Gray=1 → opposite of 1	Binary = 10
3	Gray=1 → opposite of 0	Binary = 101
4	Gray=0 → same as 1	Binary = 1011